ELSEVIER

Tectonophysics 301 (1999) 133–144

TECTONOPHYSICS

Cretaceous palaeomagnetic results from Hainan Island in south China supporting the extrusion model of Southeast Asia

Yu-Yan Liu [a], Hayao Morinaga [b,*]

[a] *Faculty of Earth Sciences, China University of Geosciences (Wuhan), Yujiashan, Wuhan 430074, People's Republic of China*
[b] *Department of Geology, Faculty of Science, Himeji Institute of Technology, Shosha 2167, Himeji 671-22, Japan*

Received 10 June 1997; accepted 18 August 1998

Abstract

We report the Cretaceous palaeomagnetic results from Hainan Island, south China. In Hainan island we collected the Early Cretaceous redbeds of the Lumuwan Formation at eleven sites. We also describe the tectonic kinematics for and around Hainan Island since the Cretaceous, deduced from our and previous palaeomagnetic results. The palaeolatitude of Hainan Island is 25.9ºN (+3.4º/−3.2º), implying that Hainan island was situated about 7º north from the present position during the Cretaceous. The palaeopole of Hainan Island (latitude = 77.7ºN, longitude = 162.1ºE, k = 65.6, and A_{95} = 4.4º) suggests 4.0 ± 5.8º counterclockwise rotation and 14.1 ± 5.5º southward translation relative to the suspected coherent part of the south China block (SCB) since the Cretaceous. The rotation and translation of similar sense (18.8 ± 7.4º and 7.8 ± 6.9º, respectively) are detected in the existing palaeomagnetic result from the Xinlong Formation in Guangxi, which is situated approximately 400 km north-northwest from Hainan Island. The southward translation of both areas seems to have been due to the southeastward extrusion of dissected zones within the southwestern part of the SCB in a similar pattern to the Indochina block, which had resulted from the indentation of India into Asia. This SW part seems to have slightly rotated counterclockwise, because its extrusion was probably smaller in scale than the Indochina block and therefore it was dragged out by the Indochina block. This hypothesis is supported by the existence of a northwest–southeast-trending fault system parallel to the Red River Fault. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: palaeomagnetism; Cretaceous; south China block; redbeds; India–Asia collision; extrusion model

1. Introduction

The south China block (SCB) located in southeast China is divided into two major fragments, one is the Yangtze craton and the other is an exotic terrane named the Huanan block. Although several different interpretations on the timing of unity of two fragments are presented (Zhang et al., 1984; Wang and Qiao, 1984; Ji and Coney, 1985; Hsu et al., 1988, 1989a,b, 1990; Rowley et al., 1989; Gupta, 1989), the unity is considered to have been completed by the middle Mesozoic at the latest (Hsu et al., 1990).

Since the Mesozoic, East and Southeast Asia have undergone tectonic deformation due to the collision of India with Asia (e.g., Tapponnier et al., 1986). Palaeomagnetic studies demonstrate tectonic movement of several blocks: southeastward translation

* Corresponding author. Tel.: +81 0792 66 1661; Fax: +81 0792 66 8868; E-mail: magnet@sci.himeji-tech.ac.jp

0040-1951/99/$ – see front matter © 1999 Elsevier Science B.V. All rights reserved.
PII: S0040-1951(98)00216-9

and clockwise rotation of the Indochina block (Yang and Besse, 1993; Yang et al., 1995), clockwise rotation of the Simao block (Huang and Opdyke, 1993), and clockwise rotation near the Indus–Zangbo suture (Otofuji et al., 1990) and near the Jinsha suture–Red River Fault (Funahara et al., 1992, 1993).

The SCB also appears to have undergone tectonic deformation due to the India–Asia collision and sea-floor spreading of the south China Sea (Briais et al., 1993). The streaked distribution of palaeopoles implies the possibility of relative rotation of small blocks within the SCB, which are dissected by faults, due to these tectonic events as well as tectonic setting (strike-slip activity) during the Mesozoic or later (Enkin et al., 1992; Gilder et al., 1993; Li et al., 1995).

Hainan Island is situated at the southernmost part of the SCB. An existing palaeomagnetic study of redbeds from the Cretaceous sedimentary basins in Hainan Island demonstrates ~10° counterclockwise rotation of Hainan Island since the Cretaceous (Li et al., 1995). Li et al. (1995) interpret that the rotation has been a regional block rotation caused by the left-lateral movement along the Red River Fault (RRF). Mo and Shi (1987) obtained a higher palaeolatitude than the present value from palaeomagnetism of the Cretaceous redbeds, and interpret that Hainan Island has translated southward by the late Palaeogene since the Cretaceous. However, we can not use the palaeomagnetic results from Mo and Shi (1987), because details of magnetic stability of the samples and field test (e.g. fold test) are not described.

This paper reports palaeomagnetic results from eleven sites of the Lower Cretaceous redbeds distributed in the central part of Hainan Island. The tectonic kinematics for and around Hainan Island since the Cretaceous is also discussed, which is deduced from the combined palaeomagnetic results of both the present study and Li et al. (1995).

2. Sampling and laboratory procedures

Redbeds often including conglomerate, named the Lumuwan Formation, are distributed in the central part of Hainan Island. The Lumuwan Formation is over 2600 m thick and exposed in generally NE-trending basins (Fig. 1). The age of the upper Lumuwan Formation is late Early Cretaceous, as determined from ostracoda and charophyte fossils (Zhang et al., 1992).

We collected palaeomagnetic samples from eleven sites in two areas of the village of Baisha: eight sites in the Nankai (N) area (19°05′N,

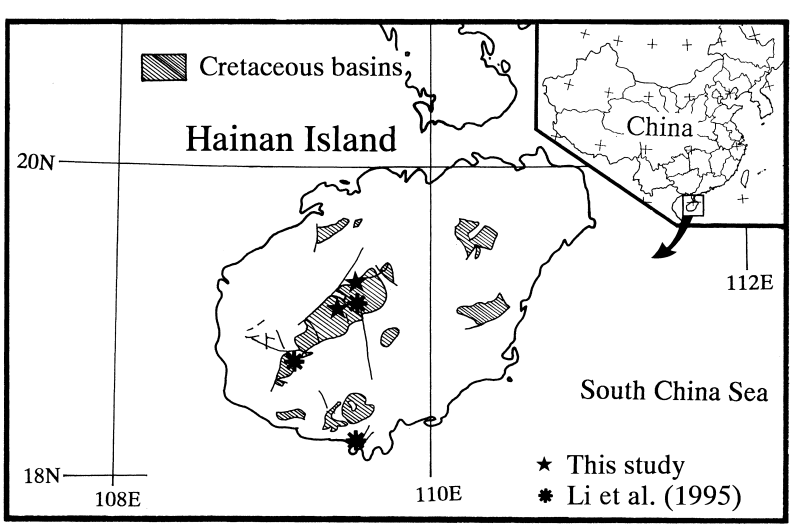


Fig. 1. Regional map of Hainan Island with sampling localities (stars) indicated. Solid lines indicate the faults noted in the Geological Map of China (Ministry of Geology and Mineral Resources of China, 1989). Asterisks indicate sampling localities of the previous study (Li et al., 1995).

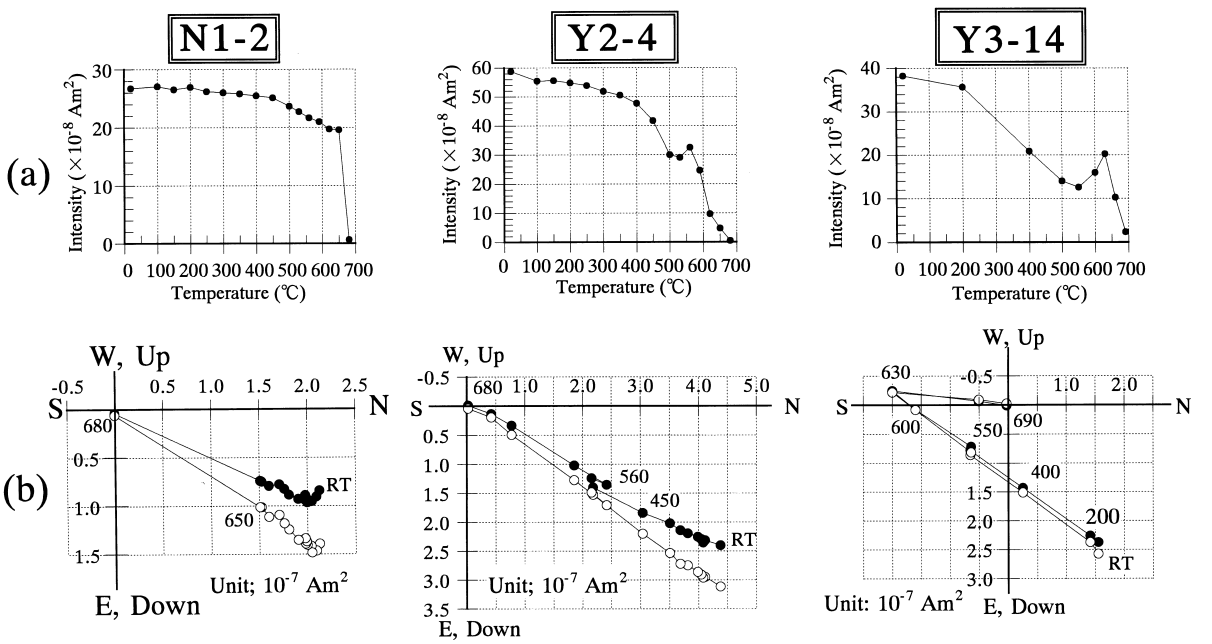


Fig. 2. Representative plots of progressive thermal demagnetization results: (a) intensity decay curves during heating, (b) orthogonal vector projections onto the horizontal plane (filled circles) and the N–S vertical plane (open circles). Numbers next to symbols indicate temperature steps (ºC). In-situ coordinates.

109º25′E) and three sites in the Yacha (Y) area (19º15′N, 109º28′E). Samples from area N are purple siltstones. Samples from area Y are red sandstones rarely containing a few mm-sized pebbles. Ten core specimens per site were collected on the average, using a portable gasoline-powered drill, and were oriented with a magnetic compass.

Remanent magnetization of core specimens was measured on a Natsuhara Giken Co. SMM-85 spinner magnetometer. Progressive thermal (Th) demagnetization was adopted. The core specimens were generally demagnetized at sixteen steps: every 50ºC from 100ºC to 500ºC and every 30ºC from 530ºC to 680ºC (or 690ºC). The characteristic direction was analyzed using the principal component analysis (Kirschvink, 1980).

3. Results of magnetic measurements and interpretation

Specimens from both the N and Y areas had a slight viscous component eliminated at low temperature steps up to 200ºC and showed a stable behaviour afterwards (Fig. 2). All the specimens from area N had only one stable component with an unblocking temperature of ∼680ºC (Fig. 2, left). The main remanence carrier appears to be hematite judging from this high unblocking temperature. We regarded this component as the characteristic remanent magnetization (ChRM) for area N specimens.

Almost all specimens from area Y showed two stable magnetization phases with different unblocking temperatures of ∼580ºC (lower temperature component = LTC) and ∼680ºC (higher temperature component = HTC), as indicated by intensity decay (Fig. 2, centre). The magnetic carriers of two phases seem to be magnetite and hematite, respectively, judging from each unblocking temperature. The directions of the two phases, however, were almost equal to each other and were not clearly separated by the principal component analysis. We regarded the HTC direction as the ChRM direction for these specimens.

Only three specimens from site Y3 showed a reversed HTC when they were demagnetized at tem-

perature steps over ~630ºC (Fig. 2, right). Li et al. (1995) also document the existence of the reversed HTCs with the same unblocking temperature. Directions of their normal LTCs are parallel to those of the LTCs and HTCs for the rest specimens from site Y3. We regarded the reversed HTC as the characteristic component for these three specimens. These three and the remaining specimens of site Y3 having only a normal component were classified as sites Y3R and Y3N, respectively.

The existence of dual polarities within one specimen gives us a hint on the duration needed to make the remanent magnetization complete. This suggests that the remanence acquisition for the redbeds was complete in much longer time than required for polarity transition. The acquisition of redbed remanence is thought to be penecontemporaneous with sedimentation, from the extensive studies of the redbeds of the Siwalik system of Pakistan (Johnson et al., 1982, 1985; Opdyke et al., 1982; Tauxe and Opdyke, 1982). Fine-grained redbeds tend to have chemical remanent magnetization (CRM) dominated by fine-grained authigenic hematite (Opdyke and Channell, 1996). Tauxe et al. (1980) demonstrated that the CRM of the redbeds of the Siwalik system was acquired early in their history. Such previous studies demonstrate that the redbed remanence is carried by two distinct phases and one of them (CRM) is certainly of post-depositional origin. The discovery of the dual polarities is convincing evidence for two distinct phases and the post-depositional acquisition of redbed remanence. Although the existence of reversed polarity specimens is significant, their data were rejected when calculating the global mean and discussing tectonic evolution of the SCB because of the small number of specimens used (3) and the low concentration ($k = 30.3$).

4. Significance test (the fold test)

Maximum concentration of the global mean for all the results including data of Li et al. (1995) is achieved at 54% untilting (Fig. 3). Li et al. (1995) have already demonstrated that the incomplete untilting (73%) brings the highest concentration. This observation again suggests that their remanences were not acquired immediately after sedimentation; they

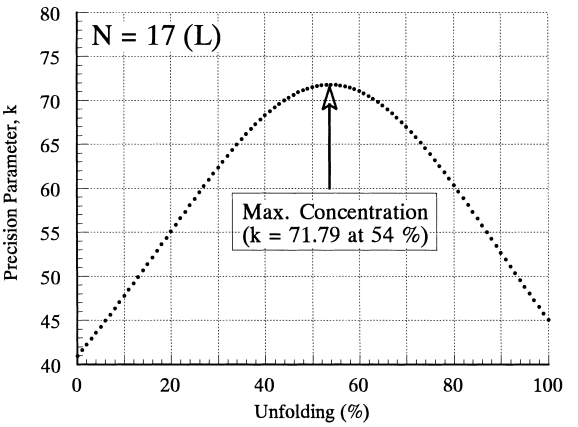


Fig. 3. Change of precision parameter (k) against degree of untilting. Maximum concentration is achieved at 54% untilting.

were acquired during tilting of the sediment layer (synfolding remanence). Most Cretaceous sedimentary basins in southeast China were controlled by fault movement (half graben or pull-apart basin). Most tilting (folding) in such tectonic basins is thought to have progressed contemporaneously with the structural and stratigraphic development of the basins (Miyata, 1990; Schlische, 1992). It is reasonable to suppose that remanence acquisition of redbeds occurred during the synsedimentary tilting, because the acquisition appears to have been post-depositional or early in the history of redbeds as mentioned above. From the existence of reversed polarity, we believe that the remanence of the Early Cretaceous redbeds from Hainan Island was acquired before the Cretaceous Long Normal Superchron (CLNS, 125 Ma).

Table 1 shows all the palaeomagnetic results and Fig. 4 shows site means of in-situ 54% and 100% untilted directions, including the results from six sites of Li et al. (1995). The precision parameter of the global mean changes from 41.0 (in situ, k_1) to 71.8 (54% untilted, k_2) and 45.1 (100% untilted, k_3). The increase of concentration ($k_2/k_1 = 1.75$) is slightly lower than a significant value (1.81 at $N = 17$) at the 95% confidence level (McElhinny, 1964). We adopt the 54% untilted directions as the palaeofield directions during the Early Cretaceous, on the basis of the discussion mentioned above: the global mean direction without Y3R ($N = 17$); dec-

Table 1
Characteristic directions for the Cretaceous from the Lumuwan Formation of Hainan Island; N and Y series are from the present study, and 1 to 4, 6 and 7 are from Li et al. (1995)

Site	Locality: lat. (°N)/lon. (°E)	Bedding strike/dip (°)/(°)	Number [a]	In-situ mean		54% untilted mean		100% untilted mean		k	α_{95} (°)	VGP calculated from 54% untilted mean: lat. (°N)/lon. (°E)	Palaeolat. (°)
				declination (°)	inclination (°)	declination (°)	inclination (°)	declination (°)	inclination (°)				
N1	19.1/109.4	171.5/14.0	10N	21.4	39.3	15.5	42.7	9.7	45.1	97.7	4.9	74.6/175.1	
N2	19.1/109.4	177.5/22.5	10N	28.3	37.9	18.9	43.3	9.0	46.2	83.5	5.3	71.5/176.7	
N5	19.1/109.4	127.5/22.5	10N	13.3	39.9	7.7	50.7	359.4	59.4	115.3	4.5	75.9/137.3	
N8	19.1/109.4	148.5/25.0	9N	16.7	41.7	6.3	51.0	352.7	56.9	103.9	5.1	76.2/132.4	
N10	19.1/109.4	134.0/15.5	10N	8.7	41.1	3.7	47.7	357.8	52.9	54.3	6.6	79.7/127.9	
N11	19.1/109.4	163.5/20.0	10N	13.9	35.7	6.3	40.6	358.4	43.5	82.9	5.3	82.8/163.4	
N13	19.1/109.4	155.5/23.5	10N	5.1	39.7	354.5	45.0	343.5	47.5	114.5	4.5	81.0/76.3	
N14	19.1/109.4	88.5/6.5	10N	9.8	41.2	10.5	44.6	11.1	47.6	95.0	5.0	78.0/161.0	
Y1	19.3/109.5	121.5/32.0	10N (HTC)	32.3	32.6	32.5	49.9	33.1	64.6	327.5	2.7	58.6/171.8	
Y2	19.3/109.5	126.0/19.5	10N (HTC)	12.0	21.1	9.8	30.6	7.0	38.6	70.6	5.8	80.3/214.8	
Y3N	19.3/109.5	122.5/27.5	9N (HTC)	23.2	36.7	20.5	51.3	15.6	63.6	354.8	2.7	67.7/160.9	
Y3R	19.3/109.5	122.5/27.5	3R (HTC)	182.3	9.4	182.7	−3.5	181.7	−14.4	17.6	30.3	−72.3/100.6	
1	19.1/109.5	167.9/34.5	10N	21.7	41.0	4.6	49.1	345.6	51.1	47.8	7.1	78.3/129.6	
2	19.1/109.5	191.0/32.5	8N	30.7	42.0	14.0	45.5	359.0	44.3	64.1	7.0	74.9/165.5	
3	19.3/109.5	172.0/28.0	8N	33.5	31.0	24.8	40.2	14.0	46.2	78.5	6.3	66.6/186.3	
4	18.3/109.5	291.0/31.0	15N	341.5	60.5	354.0	46.4	359.0	33.4	92.2	4.0	79.1/80.2	
6	18.8/109.2	194.7/10.0	6N	15.0	31.8	11.7	31.7	8.8	31.3	94.9	6.9	78.8/205.7	
7	18.8/109.2	59.0/26.7	8N	5.6	22.0	10.4	33.2	17.1	42.0	69.3	6.7	80.1/201.5	
Mean (without Y3R)			17S	16.8	37.9					$k_1 = 41.0$	5.6		
			17S			10.9	44.1			$k_2 = 71.8$	4.2	77.7/162.1 ($k = 65.6$, $A_{95} = 4.4°$)	25.9 (+3.4 ± 3.2)
			17S					4.3	48.4	$k_3 = 45.1$ $k_2/k_1 = 1.75$ $k_3/k_1 = 1.10$	5.4		

[a] N = normal polarity; R = reversed polarity; S = site; HTC = high temperature component.

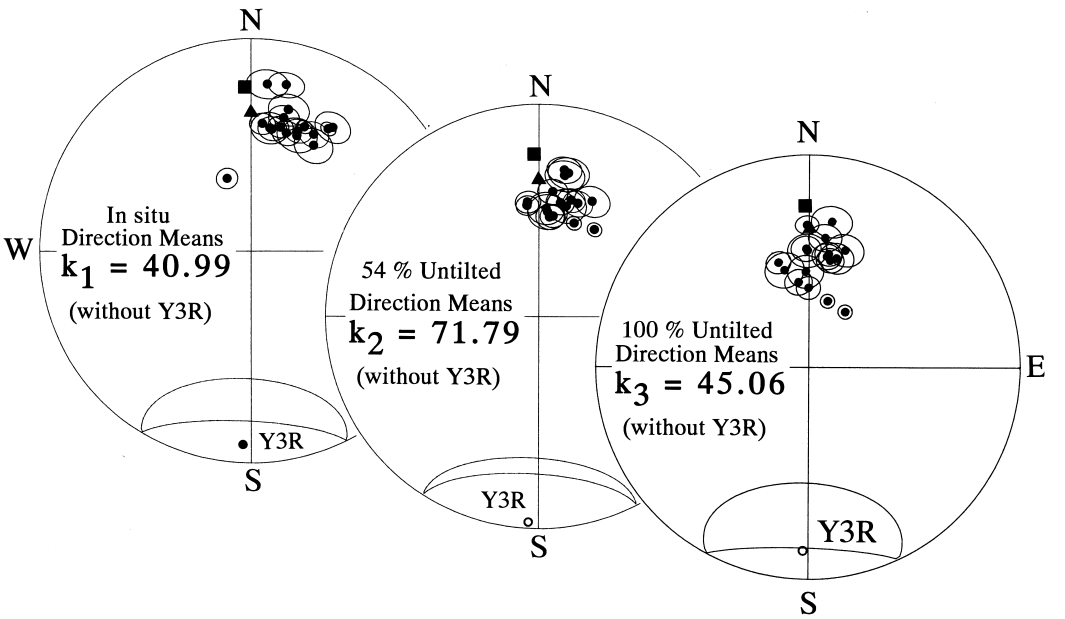


Fig. 4. Equal-area projection of all the site mean directions for the Cretaceous Lumuwan Formation; in situ (left), 54% (centre) and 100% (right) untilted directions.

lination = 10.9° and inclination = 44.1° (k = 71.8, α_{95} = 4.2°), VGP; latitude = 77.7° and longitude = 162.1° (k = 65.6, A_{95} = 4.4°), and palaeolatitude = 25.9° (+3.4°/−3.2°).

5. Southward translation of Hainan Island

Fig. 5 shows previously reported Cretaceous palaeopoles of the SCB and outside of the SCB near the RRF (Table 2). The previous palaeopoles satisfy the criteria of Enkin et al. (1992), or are obtained using mean directions of several sites, which were published recently. The streaked distribution of palaeopoles is not clear except for eight palaeopoles for central Yunnan (No. 1 in Fig. 5) (Funahara et al., 1992), near the Baise Fault in Guangxi (No. 2) (Gilder et al., 1993), coastal regions east or south of the SCB (No. 3 from Jiangsu (Kent et al., 1986), No. 4 from Hong Kong (Chan, 1991), No. 5 from Hainan Island (Li et al., 1995, and this study)), and Nos. 12, 13, 14, 15 and 16 from outside of the SCB (Yongping in Yunnan (Funahara et al., 1993), Khorat in Thailand (Yang and Besse, 1993), Jinggu, Mengla and Xiaguan in Yunnan (Huang and Opdyke, 1993)). The remaining six SCB palaeopoles (Nos. 6, 7, 8 and 9 in Fig. 5 from Sichuan (Enkin et al., 1991a,b; Huang and Opdyke, 1992), No. 10 from Zhejiang (Lin, 1987) and No. 11 from Fujian (Zhai et al., 1992)) are located at almost equal positions. This indicates that their areas have neither rotated nor translated relative to one other, and implies that a region covering these sampling areas has behaved as one coherent block since the Cretaceous.

The palaeopole for Hainan Island (No. 5) is located at 14.1 ± 5.5° near-sided position as compared with the mean of six palaeopoles on the suspected coherent part of the SCB (latitude = 74.0°N, longitude = 223.3°E, k = 158.7 and A_{95} = 5.3°). This implies that Hainan Island has translated southward about 14° relative to the coherent part of the SCB since the Cretaceous. Hainan Island appears to have been also slightly rotated counterclockwise by 4.0±5.8° since the Cretaceous. The palaeolatitude of Hainan Island at the Early Cretaceous is calculated at 25.9° (+3.4°/−3.2°) from the mean inclination and is higher by approximately 7° than the present latitude (about 19°).

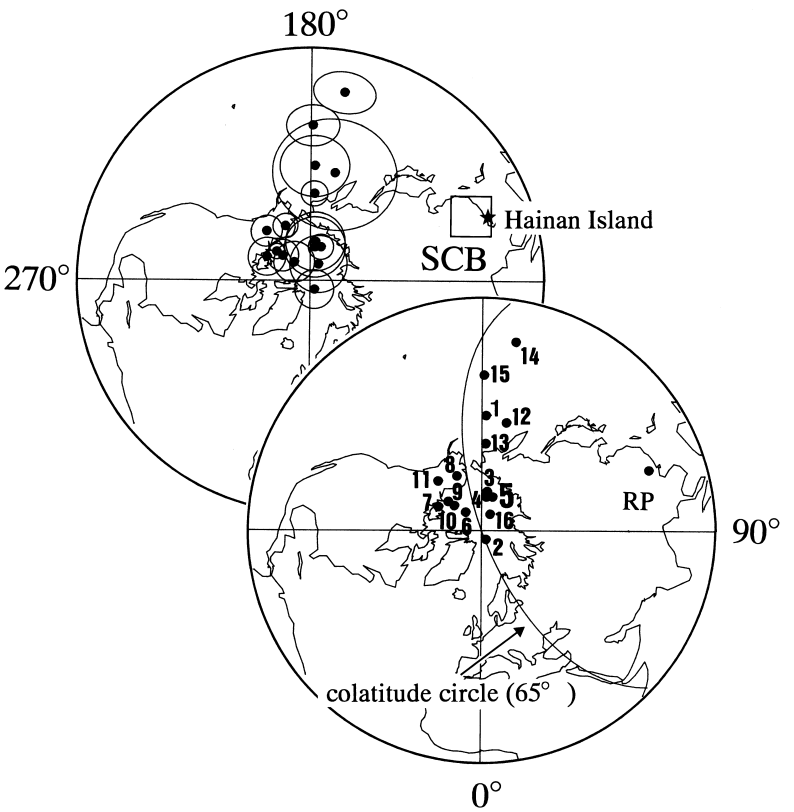


Fig. 5. Cretaceous palaeopoles from the south China block and outside of the SCB near the RRF: *1* from Yunnan (Funahara et al., 1992); *2* from Guangxi (Gilder et al., 1993); *3* from Jiangsu (Kent et al., 1986); *4* from Hong Kong (Chan, 1991); *5* from Hainan Island (Li et al., 1995 and this study); *6, 7, 8* and *9* from Sichuan (Enkin et al., 1991a,b; Huang and Opdyke, 1992); *10* from Zhejiang (Lin, 1987); *11* from Fujian provinces (Zhai et al., 1992); *12* from Yunnan east of the Red River Fault (Funahara et al., 1993); *13* from Khorat in Indochina (Yang and Besse, 1993); *14, 15* and *16* from Jinggu, Mengla and Xiaguan in Yunnan, respectively (Huang and Opdyke, 1993). The small circle in the bottom figure indicates a colatitude circle of 65º from the reference point of the SCB (25ºN and 110ºE, RP). This line gives tentative criteria to judge which palaeopoles are near-sided or far-sided from the SCB.

6. Tectonic kinematics around Hainan Island

The tendency of southward translation relative to the suspected coherent part of the SCB is recognized with eight Cretaceous palaeopoles for Yunnan (Nos. 1, 12, 14 and 15 in Fig. 5), Guangxi (No. 2), Jiangsu (No. 3), Hong Kong (No. 4), and Indochina (No. 13) (bottom part of Fig. 5). Among these eight, the palaeopole for Guangxi (No. 2, Gilder et al., 1993) is located at the position near-sided by $7.8 \pm 6.9^{\circ}$ and implying the $18.8 \pm 7.4^{\circ}$ counterclockwise rotation of the area, relative to the suspected coherent part of the SCB as well as Hainan Island. This similar sense of translation and rotation suggests that both areas have undergone a common tectonic evolution. We regard the tectonic evolution as the southeastward extrusion of a region including both areas.

A similar magnitude of southward translation is recognized for the Khorat basin of the Indochina block (No. 13 in Fig. 5) and Yongping, Jinggu, Mengla and Xiaguan areas in the Simao block (Nos. 12, 14, 15 and 16). The southward translations are calculated at $13.7 \pm 5.6^{\circ}$, $17.3 \pm 17.0^{\circ}$, $11.7 \pm 8.3^{\circ}$, $8.9 \pm 7.8^{\circ}$ and $13.1 \pm 9.6^{\circ}$, respectively, relative to the suspected coherent part of the SCB, using the existing Cretaceous palaeomagnetic results (Table 2; Yang and Besse, 1993; Funahara et al., 1993; Huang and Opdyke, 1993).

Fig. 6 shows the faults noted in the geological map of the southern part of China (Ministry of Ge-

Table 2
Cretaceous palaeodirections and palaeopoles from the SCB and outside of the SCB near the Red River Fault

Age	Locality: (lat °N/lon °E)	No. of sites	Direction: decli/incli	Precision α_{95}	Palaeolat	VGP: lat(N)/lon(E)	Precision A_{95}	Pole no. in Fig. 5	Reference
SCB									
K2	Nanjing, Jiangsu (32.0/119.0)	10	13.7/58.0	7.6	38.7	76.3/172.6	10.3	3	Kent et al., 1986
K2	Xiaoba, Sichuan (26.5/102.4)	18	8.1/38.8	6.6	21.9	81.5/220.9	7.1	6	Huang and Opdyke, 1992
K2	Yongan, Fujian (26.0/117.3)	5	23.5/33.5	7.3	18.3	66.9/221.4	5.4	11	Zhai et al., 1992
K2	Feixianguan, Sichuan (30.0/102.9)	16	–	–	–	72.8/241.1	6.6	7	Enkin et al., 1991a,b
K	Chuxiong, Yunnan (25.0/101.5)	21	44.6/41.3	10.7	23.7	49.2/178.0	11.4	1	Funahara et al., 1992
K1	Hong Kong (22.3/114.2)	12	11.5/47.8	8.7	28.9	78.2/171.9	10.6	4	Chan, 1991
K1	Feitianshan, Sichuan (26.8/102.5)	7	22.1/37.1	3.8	20.7	69.0/204.6	4.3	8	Huang and Opdyke, 1992
K1	Zhejiang (29.6/120.5)	7	13.6/42.7	6.1	24.8	77.1/227.6	5.5	10	Lin, 1987
K1	Xinlong, Guangxi (22.7/108.7)	8	352.8/38.2	5.5	21.5	86.5/26.4	6.8	2	Gilder et al., 1993
K1 (K2–J3)	Feixianguan, Sichuan (30.0/102.9)	23	–	–	–	74.5/229.0	2.7	9	Enkin et al., 1991a
K1	Hainan Island (19.1/109.4)	17	10.9/44.1	4.2	25.9	77.7/162.1	4.4	5	Li et al., 1995 + this study
Outside of SCB (near RRF)									
K	Yongping, Yunnan (25.5/99.5)	12	42.0/51.1	15.7	31.8	50.9/167.3	20.6	12	Funahara et al., 1993
K	Khorat, Thailand (16.7/101.8)	10	32.0/38.2	4.1	21.5	59.4/177.4	4.5	13	Yang et al., 1995
K	Jinggu, Yunnan (23.4/100.9)	8	79.4/43.3	9.1	25.2	18.9/170.0	8.9	14	Huang and Opdyke, 1993
K	Mengla, Yunnan (21.6/101.4)	10	60.8/37.8	7.6	21.2	33.7/179.3	8.2	15	Huang and Opdyke, 1993
K	Xiaguan, Yunnan (25.6/100.2)	9	6.9/47.7	8.6	28.8	83.6/152.7	10.0	16	Huang and Opdyke, 1993

ology and Mineral Resources of China, 1989). The sampling area (the Xinlong Formation) for Guangxi (No. 2 in Fig. 6) is situated approximately 400 km from Hainan Island. The sampling localities for Guangxi and Hainan Island (No. 5 in Fig. 6) are at a less than 400 km distance northeast the RRF or the suspected extension of the RRF (A in Fig. 6, the boundary between the SCB and Indochina block). Several NW–SE-trending faults parallel to the RRF are present in this region. The Baise Fault Zone (C in Fig. 6) is one of the NW–SE-trending faults. Within the northeastern part of the SCB more than 400 km from the RRF, NE–SW-trending faults are dominant, considered to have been active from the Triassic to the Early Cretaceous (Xu et al., 1987).

Fig. 7 illustrates the southeastward extrusion model of the southwestern part of the SCB. The Indochina block (hatched in Fig. 7) extruded southeastward by the stress resulted from the indentation of India into Asia (Yang and Besse, 1993; Yang et al., 1995). The Simao block may have also experienced the southeastward or southward extrusion by the stress, because the southward translation is inferred palaeomagnetically. The stress probably extended to the southwest part (shaded in Fig. 7) of SCB over the RRF and dissected the southwest part by several

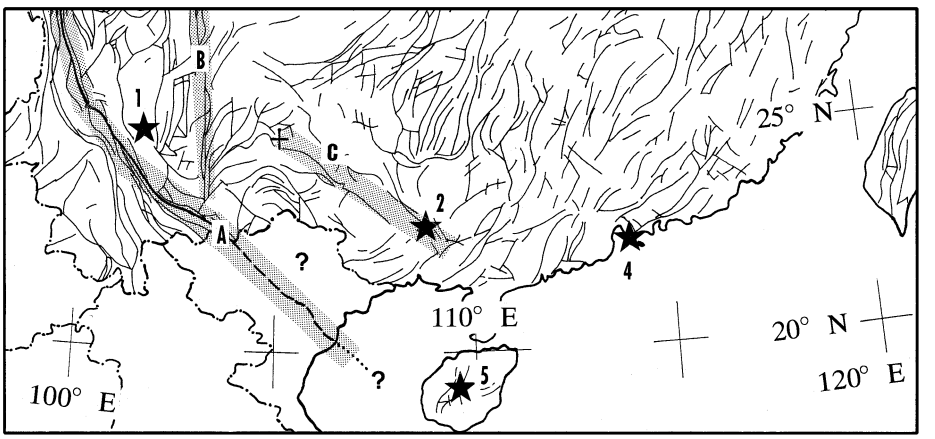


Fig. 6. Tectonic map of the south China. Solid lines indicate the faults noted in the Geological Map of China (Ministry of Geology and Mineral Resources of China, 1989): *A* = Red River Fault zone, *B* = Xianshuihe–Xiaojiang Fault zone, and *C* = Baise Fault zone. Stars indicate the sampling localities for Cretaceous palaeomagnetic studies in the SCB; *1* = Chuxiong of the central Yunnan (Funahara et al., 1992); *2* = Xinlong, Guangxi (Gilder et al., 1993); *4* = Hong Kong (Chan, 1991); *5* = Hainan Island (Li et al., 1995 and this study).

faults parallel to the RRF. The dissected zones nearer to the RRF were extruded on a larger scale than the zones further distant from the RRF. This corresponds to the observation of the larger southward translation for Hainan Island (about 14º) than that for Guangxi (about 8º). The extrusion of the southwest part of the SCB was of a smaller scale by at least 500 km than that of the Indochina block, because at least 500 km of sinistral motion has occurred along the RRF, of which Leloup et al. (1995) give the geological evidence. Therefore the southwest part was probably dragged out by the extrusion of the Indochina block and then rotated counterclockwise by a small amount (about 4º and 19º for Hainan Island and Guangxi, respectively). As the stress was probably absorbed within the southwest part of the SCB, it did not extend over the northeast part of the SCB and therefore faults parallel to the RRF decrease in number gradually with distance from the RRF.

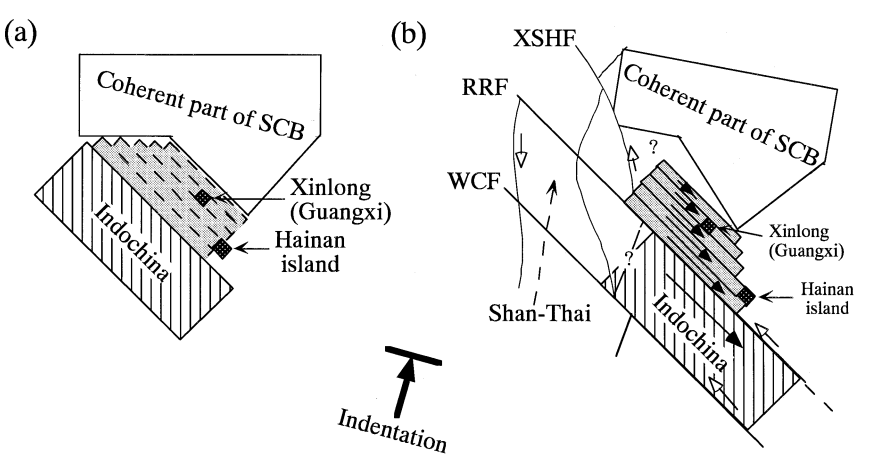


Fig. 7. Scheme on southeastward extrusion of the southwest part (shaded) of the SCB resulted from the indentation of India into Asia; (a) before the indentation (the Early Cretaceous) and (b) the latest stage of extrusion. Solid arrows correspond to the direction and scale of extrusion described in the text, and open ones correspond to the sense of strike-slip motion on major faults for older phases after Yang et al. (1995); *RRF* = Red River Fault, *XSHF* = Xianshuihe–Xiaojiang Fault and *WCF* = Wang Chao Fault.

The tectonic kinematics mentioned above is one hypothesis based only on palaeomagnetic results. The southward translation of 14° calculated for Hainan Island corresponds to approximately 2200 km southeastward motion of Hainan Island. This motion may be rather overestimated, because major tectonic features suggesting the large southeastward motion have not been documented within the southwest part of the SCB at present. Gilder et al. (1993) document that the Youjiang Fault in Guangxi has a smaller sinistral motion (24 ± 7 km) inferred palaeomagnetically. Even if such a displacement estimated on both sides of one fault is small, this inference gives the possibility that the existence of several faults resulted in larger displacements against the coherent part of the SCB.

The palaeopole for central Yunnan within the SCB (No. 1 in Fig. 5) is located at a position different from those for Guangxi and Hainan Island, showing the clockwise rotation of the region. The clockwise rotation is not explained by the southeastward extrusion as well as Guangxi and Hainan Island. Central Yunnan appears to belong to a different tectonic zone than Guangxi and Hainan Island as separated by the north–south trending faults (Xianshuihe–Xiaojiang Fault zone, B in Fig. 6). The near-sided distribution of palaeopoles from Hong Kong (No. 4 in Figs. 5 and 6) and Jiangsu (No. 3 in Fig. 5) in the coastal region is not explained by movement of northeast–southwest-trending left-lateral faults in the Mesozoic (Xu et al., 1987). These unsolved questions show the necessity of further Cretaceous palaeomagnetic investigations for the central Yunnan and the coastal region.

7. Conclusion

The Early Cretaceous palaeomagnetic results from Hainan Island demonstrate the following tectonic kinematics for and around Hainan Island since the Cretaceous, which supports the extrusion model of Southeast Asia (e.g., Tapponnier et al., 1986).

(1) Hainan Island was situated at about a 7° higher latitude than the present position (about 19°N) during the Early Cretaceous.

(2) Hainan Island rotated counterclockwise by $4.0 \pm 5.8°$ and translated southward by $14.1 \pm 5.5°$. The rotation and translation of similar sense are detected also in the existing palaeomagnetic results from the Xinlong Formation (Guangxi). The rotation and translation is calculated at $18.8 \pm 7.4°$ counterclockwise and $7.8 \pm 6.9°$ southward, respectively.

(3) The southward translation of both areas was due to the southeastward extrusion of dissected zones within the southwestern part of the SCB, which resulted from the India–Asia collision. This demonstration is supported by the existence of several northwest–southeast-trending faults parallel to the Red River Fault.

(4) The southwest part of the SCB was extruded southeastwards on a smaller scale than the Indochina block. As a result, this part was dragged out by the southeastward extrusion of the Indochina block and therefore rotated counterclockwise by a small amount.

Acknowledgements

We thank Prof. Yang Shennan of China University of Geosciences (Wuhan) who financially supported this study, and Mr. Lu Lianzhong for assistance during the field work in Hainan Island. We also appreciate informative and critical comments by Drs. Stuart A. Gilder, Randolph J. Enkin and an anonymous person as reviewers of the first and/or second manuscripts and by Dr. T. Miyata of Kobe University. This study was partly supported by Grant-in-Aid for International Scientific Research (University-to-University Cooperative Research) from the Japanese Ministry of Education, Science, Sports and Culture (No. 07045014, to Dr. T. Miyata of Kobe University).

References

Briais, A., Patriat, P., Tapponnier, P., 1993. Updated interpretation of magnetic anomalies and seafloor spreading stages in the South China Sea: implications for the Tertiary tectonics of southeastern Asia. J. Geophys. Res. 98, 6299–6328.

Chan, L., 1991. Paleomagnetism of late Mesozoic granitic intrusions in Hong Kong: implications for upper Cretaceous reference pole of South China. J. Geophys. Res. 96, 327–335.

Enkin, R., Chen, Y., Courtillot, V., Besse, J., Xing, L., Zhang, Z., Zhuang, Z., Zhang, J., 1991a. A Cretaceous pole from south

China and the Mesozoic hairpin turn of the Eurasian apparent polar wander path. J. Geophys. Res. 96, 4007–4027.

Enkin, R., Courtillot, V., Xing, L., Zhang, Z., Zhuang, Z., Zhang, J., 1991b. The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block). Tectonics 10, 547–559.

Enkin, R., Yang, Z., Chen, Y., Courtillot, V., 1992. Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the Present. J. Geophys. Res. 97, 13953–13989.

Funahara, S., Nishiwaki, N., Miki, M., Murata, F., Otofuji, Y., Wang, Y., 1992. Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China, implications for the India–Asia collision. Earth Planet. Sci. Lett. 113, 77–91.

Funahara, S., Nishiwaki, N., Murata, F., Otofuji, Y., Wang, Y., 1993. Clockwise rotation of the Red River fault inferred from paleomagnetic study of Cretaceous rocks in the Shan-Thai-Malay block of western Yunnan, China. Earth Planet. Sci. Lett. 117, 29–42.

Gilder, S., Coe, R., Wu, H., Kuang, G., Zhao, X., Wu, Q., Tang, X., 1993. Cretaceous and Tertiary paleomagnetic results from Southeast China and their tectonic implications. Earth Planet. Sci. Lett. 117, 637–652.

Gupta, S., 1989. Comments on 'Mesozoic over thrust tectonics in south China'. Geology 17, 669–671.

Hsu, K., Sun, S., Li, J., Chen, H., Pen, H., Sengor, A., 1988. Mesozoic over thrust tectonics in south China. Geology 16, 418–421.

Hsu, K., Sun, S., Li, J., Chen, H., Pen, H., Sengor, A., 1989a. Reply to Rowley et al. (1989) on 'Mesozoic over thrust tectonics in south China'. Geology 17, 386–387.

Hsu, K., Sun, S., Li, J., Chen, H., Pen, H., Sengor, A., 1989b. Reply to Gupta (1989) on 'Mesozoic over thrust tectonics in south China'. Geology 17, 672–673.

Hsu, K., Li, J., Chen, H., Wang, Q., Sun, S., Sengor, A., 1990. Tectonics of South China, key to understanding West Pacific geology. Tectonophysics 183, 9–39.

Huang, K., Opdyke, N., 1992. Paleomagnetism of Cretaceous to lower Tertiary rocks from Southwestern Sichuan, a revisit. Earth Planet. Sci. Lett. 112, 29–40.

Huang, K., Opdyke, N., 1993. Paleomagnetic results from Cretaceous and Jurassic rocks of South and Southwest Yunnan, evidence for large clockwise rotations in the Indochina and Shan-Thai-Malay terranes. Earth Planet. Sci. Lett. 117, 507–524.

Ji, X., Coney, P., 1985. Accreted terranes in China. In: Howell, D. (Ed.), Tectonostratigraphic Terranes of the Circum-Pacific Region. Circum-Pac. Counc. Energy Miner. Resour. Earth Sci., Ser. 1, pp. 349–361.

Johnson, N., Opdyke, N., Johnson, G., Lindsay, E., Tahirkheli, R., 1982. Magnetic polarity stratigraphy and ages of Siwalik Group rocks of the Potwar Plateau, Pakistan. Palaeogeogr., Palaeoclimatol., Palaeoecol. 37, 17–42.

Johnson, N., Stix, J., Tauxe, L., Cerveny, P., Tahirkheli, R., 1985. Paleomagnetic chronology, fluvial processes and tectonic implications of the Siwalik deposits near Chinji Village, Pakistan. J. Geol. 93, 27–40.

Kent, D., Xu, G., Huang, K., Zhang, W., Opdyke, N., 1986. Paleomagnetism of upper Cretaceous rocks from South China. Earth Planet. Sci. Lett. 79, 179–184.

Kirschvink, J., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Leloup, P., Lacassin, R., Tapponnier, P., Scharer, U., Zhong, D., Liu, X., Zhang, L., Ji, S., Trinh, P.T., 1995. The Ailao Shan–Red River shear zone (Yunnan, China), Tertiary transform boundary of Indochina. Tectonophysics 251, 3–84.

Li, Z., Metcalfe, I., Wang, X., 1995. Vertical-axis block rotation in southwestern China since the Cretaceous, new paleomagnetic results from Hainan Island. Geophys. Res. Lett. 22, 3071–3074.

Lin, J., 1987. The Apparent Polar Wander Paths for the North and South China Blocks. Ph.D. thesis, Univ. of Calif., Santa Barbara, 248 pp.

McElhinny, M., 1964. Statistical significance of the fold test in palaeomagnetism. Geophys. J. R. Astron. Soc. 8, 338–340.

Ministry of Geology and Mineral Resources of China, 1989. Geological Map of China (in Chinese). Geological Publishing House, Beijing.

Miyata, T., 1990. Slump strain indicative of paleoslope in Cretaceous Izumi sedimentary basin along Median tectonic line, southwest Japan. Geology 18, 392–394.

Mo, Y., Shi, Y., 1987. Paleomagnetic study and tectonic evolution of Hainan terrane and its vicinal continental coast the late Mesozoic to Cenozoic (in Chinese with English abstract). J. Nanjing Univ. (Nat. Sci. Ed.) 23, 521–532.

Opdyke, N., Channell, J., 1996. Magnetic Stratigraphy. Academic Press, San Diego, Calif., 346 pp.

Opdyke, N., Johnson, N., Johnson, G., Lindsay, E., Tahirkheli, R., 1992. Paleomagnetism of the middle Siwalik formations of northern Pakistan and rotation of the Salt Range Decollement. Palaeogeogr., Palaeoclimatol., Palaeoecol. 37, 1–15.

Otofuji, Y., Inoue, Y., Funahara, S., Murata, M., Zheng, X., 1990. Palaeomagnetic study of eastern Tibet — deformation of the Three Rivers region. Geophys. J. Int. 103, 85–94.

Rowley, D., Ziegler, A., Nie, G., 1989. Comments on 'Mesozoic over thrust tectonics in south China'. Geology 17, 384–386.

Schlische, R., 1992. Structural and stratigraphic development of the Newark extensional basin, eastern North America, evidence for the growth of the basin and its bounding structures. Geol. Soc. Am. Bull. 104, 1246–1263.

Tapponnier, P., Peltzer, G., Armijo, R., 1986. On the mechanics of the collision between India and Asia. In: Coward, M., Ries, A. (Eds.), Collision Tectonics. Geol. Soc. London Spec. Publ. 19, 115–157.

Tauxe, L., Opdyke, N., 1982. A time framework based on magnetostratigraphy for the Siwalik sediments of the Khaur area, northern Pakistan. Palaeogeogr., Palaeoclimatol., Palaeoecol. 37, 43–61.

Tauxe, L., Kent, D., Opdyke, N., 1980. Magnetic components contributing to the NRM of Middle Siwalik red beds. Earth Planet. Sci. Lett. 47, 279–284.

Wang, H., Qiao, X., 1984. Proterozoic stratigraphy and tectonic framework in China. Geol. Mag. 121, 599–614.

Xu, J., Zhu, G., Tong, W., Cui, K., Liu, Q., 1987. Formation and evolution of the Tancheng–Lujiang wrench fault system, a major shear system to the northwest of the Pacific Ocean. Tectonophysics 134, 273–310.

Yang, Z., Besse, J., 1993. Paleomagnetic study of Permian and Mesozoic sedimentary rocks from northern Thailand supports the extrusion model for Indochina. Earth Planet. Sci. Lett. 117, 525–552.

Yang, Z., Besse, J., Sutheetorn, V., Bassoullet, J.P., Fontaine, H., Buffetaut, E., 1995. Lower-Middle Jurassic paleomagnetic data from the Mae Sot area (Thailand), Paleogeographic evolution and deformation history of Southeastern Asia. Earth Planet. Sci. Lett. 136, 325–341.

Zhai, Y., Seguin, M., Zhou, Y., Dong, J., Zheng, Y., 1992. New paleomagnetic data from the Huanan block, China and Cretaceous tectonics in eastern China. Phys. Earth. Planet. Inter. 73, 163–188.

Zhang, Z., Liou, J., Coleman, R., 1984. An outline of the plate tectonics of China. Geol. Soc. Am. Bull. 95, 295–312.

Zhang, Z., Meng, F., Sheng, X., 1992. Cretaceous (in Chinese). In: Wang, X., Ma, D., Jiang, D. (Eds.), Geology of Hainan Island (I). Geological Publishing House, Beijing, pp. 200–217.